



Make Your Own Dual-Voltage Rechargeable Torch Light

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The purpose of this circuit is to create a new topology with good power factor maintained at AC voltage input, to provide good efficiency output to the battery and to increase its life.

Circuit diagram of the dual-voltage rechargeable torch light is shown in Fig. 1. It is built around two mylar 0.82μF capacitors (C1 and C2), four rectifier diodes 1N4007 (D1 through D4), two 3.7V, 860mAh lithium-ion rechargeable batteries (BATT.1 and BATT.2), two 5mm LEDs (LED1 and LED2), two one-watt LEDs (LED3 and LED4) and a few other components.

Charging of the two batteries is done via the circuit using capacitors C1 and C2 and resistors R2 and R3. Two sets of RC parallel circuits are used to step down 230V AC mains voltage before giving it to the batteries. Diodes D1 and D2 are used for rectification purposes. It is recommended to use 4.7V Zener diode across capacitors C3 and C4.

Positive half cycle of the AC mains charges BATT.1, and negative half cycle charges BATT.2. Capacitors C3 and C4 are filtering elements used to reject ripples and harmonics present in DC voltages.

PARTS LIST

Semiconductors:

- D1-D4 - 1N4007 rectifier diode
- LED1, LED2 - 5mm LED
- LED3, LED4 - 1W LED

Resistors (all 1/4-watt, ±5% carbon), unless stated otherwise:

- R1 - 1-ohm, 0.5W
- R2, R3 - 100-kilo-ohm, 0.5W
- R4, R5 - 2.2-kilo-ohm

Capacitors:

- C1, C2 - 0.82μF, 400V mylar disk
- C3, C4 - 22μF, 16V electrolytic

Miscellaneous:

- S1 - DPDT switch
- BATT.1, BATT.2 - 3.7V, 860mAh lithium-ion battery
- 2-pin terminal connector
- 230V, 50Hz AC mains power supply

(All these are available from kitsnspares.com)

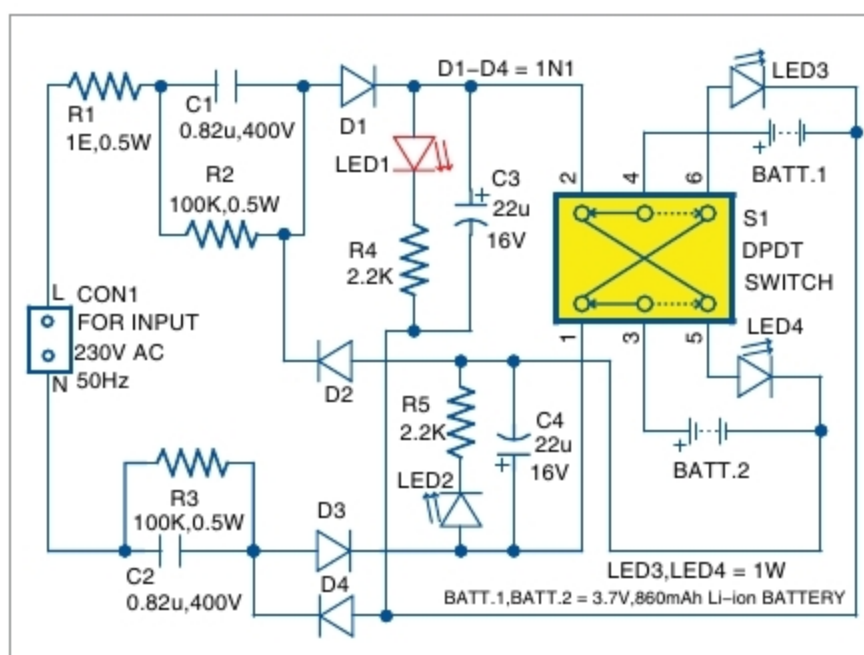


Fig. 1: Circuit diagram of the dual-voltage rechargeable torch light

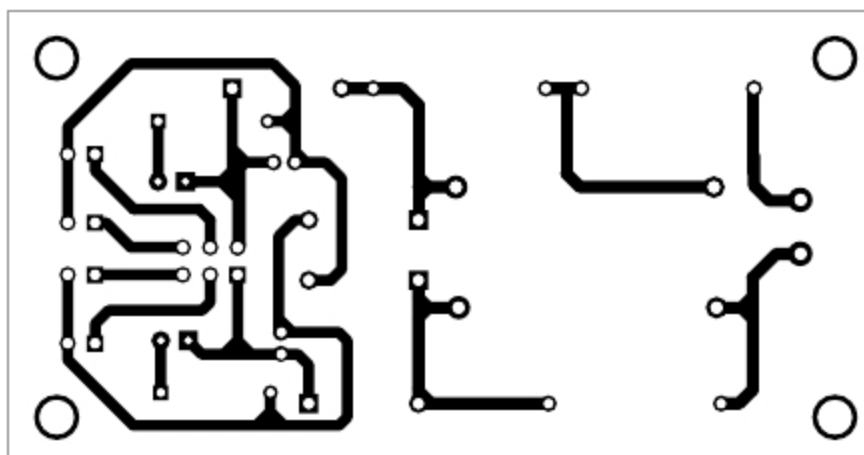


Fig. 2: Actual-size PCB layout of the rechargeable torch light

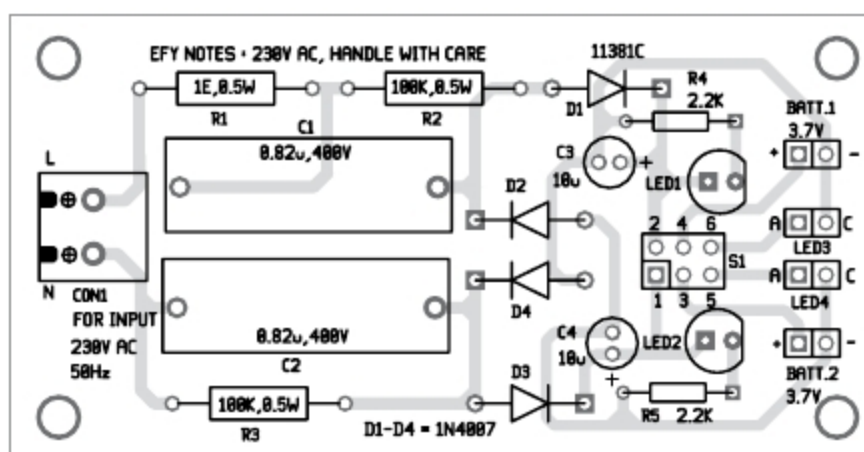


Fig. 3: Components layout for the PCB

Both LED3 and LED4 are one-watt white LEDs used for the torch application.

For charging the two batteries, keep DPDT switch S1 towards positions 1 and 2, and connect 230V AC

input across CON1. The batteries' charging status is indicated by the glowing of LED1 and LED2. After charging the batteries, the circuit can be used for the torch light. For using as a torch, keep S1 towards positions 5 and 6, so the white LEDs (LED3 and LED4) will glow.

Construction and testing

An actual-size PCB layout of the dual-voltage rechargeable torch light is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB. Connect 230V AC into the PCB across CON1. Connect LED3 and LED4 for the torch and two lithium-ion batteries for charging.

Caution. Care should be taken to avoid electric shock while handling 230V AC mains power supply. Also, it is essential to ensure that AC mains supply is switched off after charging because the circuit does not have automatic cut-off feature when the batter-

ies are fully charged. **EFY**



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